

Brain network alterations in trauma and post-traumatic stress disorder: a systematic review

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Academic Editor: Roberto Ciccocioppo

Abstract

Introduction: A substantial body of evidence suggests that traumatic experiences and post-traumatic stress disorder (PTSD) profoundly affect the brain. This systematic review investigates the neurobiological correlates of trauma by synthesizing neuroimaging findings on resting-state activity in PTSD and trauma-exposed participants. By integrating data on regional activity (metabolism/blood flow) and functional connectivity, we aim to distinguish PTSD-specific neural signatures from general trauma-related alterations. We also examine how recent emotional or cognitive engagement may modulate these resting-state patterns.

Materials and methods: The electronic databases PubMed, Scopus, and PsycINFO were systematically searched for references on brain correlates of traumatic experiences and PTSD. Target populations included patients with PTSD or individuals exposed to traumatic events without a PTSD diagnosis. References were screened, and the included studies were assessed for quality of evidence. Resting-state functional activation patterns associated with traumatic experiences and PTSD were explored.

Results: Seventeen studies were included. These studies highlighted alterations in specific brain networks, such as the default mode network, salience network, and central executive network, in individuals with traumatic experiences and PTSD. Common findings included alterations in the medial prefrontal cortex and the subcortical regions involving the amygdala and the hippocampus in individuals with PTSD.

Conclusions: This systematic review underscores the brain activation patterns associated with traumatic experiences and PTSD. The identified alterations in key brain networks provide insights into the neural underpinnings of these conditions and suggest potential targets for future therapeutic interventions. Further research with standardized methodologies and larger cohorts is warranted to confirm these findings and explore their clinical implications.

Keywords: *traumatic experience, post-traumatic stress, resting-state functional connectivity, corticolimbic circuit, default mode network, salience network*

Citation: Imtiaz S, Panico F, Catalano L, De Biase R, Sagliano L. Brain network alterations in trauma and post-traumatic stress disorder: a systematic review. *Academia Neuroscience and Brain Research* 2026;2. <https://doi.org/10.20935/AcadNeurosci8253>

1. Introduction

Traumatic experiences or exposure to stressful situations variably characterize individuals' lives. Some grievous events involving severe injury, abuse, violence, threat to the physical integrity of self, and actual or threat of death, can have a profound psychological effect with a high traumatic potential; nonetheless, many other stressful events eliciting intense emotions such as fear, helplessness, or terror may have variable effects depending on the individual's personal resources [1, 2]. Overall, these experiences could profoundly affect an individual's mental health and well-being, potentially resulting in psychiatric disorders such as post-traumatic stress disorder (PTSD) [3–5]. According to epidemiological research, about 9% of individuals who experience traumatic stressors may develop PTSD; however, this percentage varies depending on the type of trauma and the individual's characteristics [2]. The lifetime prevalence estimates of PTSD in the general adult population in North American contexts range from 6.1% to 9.2% [6–9]. According to the research conducted in a European context, the lifetime prevalence of PTSD is 7.8%; however, prevalence rates of the disorder may vary across different areas and regions [10, 11].

Previous evidence has suggested an association between exposure to trauma and alterations in the structure and function of several brain areas [12, 13]. Specifically, as the prefrontal cortex (PFC) is closely connected to the hippocampus (HIP) and amygdala (AMG), any disruption to the functional connectivity of this stress response network could directly contribute to the development of PTSD [14–16].

Various neuroimaging techniques have been employed to investigate the neural mechanisms underlying stress and trauma [13, 17]. Our study specifically focuses on the findings of various functional applications to provide a comprehensive characterization of the post-traumatic brain.

In particular, functional magnetic resonance imaging (fMRI), specifically resting-state fMRI (RS-fMRI), has become pivotal in studying the pathophysiology of PTSD. In this context, functional connectivity (FC) reflects the temporal dependence of neurophysiological indices across anatomically separated regions [18–20]. Additionally, RS-fMRI provides insights into spontaneous brain activity independent of task-based stimulation [21–23], utilizing

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indices such as regional homogeneity (ReHo) to assess local functional activity [24].

In general, brain networks are anatomically distinct but demonstrate significant functional interconnectedness within the brain regions involved, thereby substantiating sustained functional connectivity at rest [19, 25–30]. The eight identified networks include the default mode network (DMN), the salience network (SN), the central executive network (CEN), the dorsal attention network (DAN), the limbic network (LIN), the visual network (VIS), the somatomotor network (SMN), and the lateralized network (consisting of superior parietal and superior frontal regions).

The DMN comprises the posterior cingulate cortex (PCC), medial prefrontal cortex (MPFC), HIP, precuneus, and inferior parietal and lateral/medial temporal cortices [19, 20, 25, 27–31] and is activated during resting-state and self-referential processes [26, 32]. The salience network encompasses the anterior cingulate cortex (ACC), insula, AMG, and substantia nigra and is activated when processing salient extrinsic and intrinsic information (specifically emotions); therefore, it may be responsible for influencing negative affective states [33, 34]. The central executive network, also known as the frontoparietal network, includes the inferior frontal gyrus (IFG), lateral prefrontal cortex (IPFC), and dorsal anterior cingulate cortex (dACC), and is activated during executive processing, e.g., working memory or decision-making [33, 34].

According to the Menon's triple network model [33] PTSD could be associated with altered large-scale network connectivity in three of the above presented large-scale networks, namely, DMN, SN, and CEN. Moreover, disrupted functional connectivity within or between the anatomically segregated brain regions in PTSD might result in transdiagnostic symptoms (e.g., re-experiencing, avoidance, arousal, rumination, and worry) and could play a substantial role in psychopathology [33, 34]. For example, dysfunctional DMN functional connectivity could be responsible for the re-experiencing, dissociation, and avoidance symptoms in PTSD while dysfunctional connectivity of SN could be associated with hyperarousal and increased sensitivity to threat detection. Moreover, disrupted functional CEN connectivity could be associated with reduced prefrontal inhibition and attentional and memory biases [35]. Finally, aberrant hippocampal activity in PTSD is associated with biases in stimuli interpretation across cognitive, social, and internal domains, and might determine the failure to interpret trauma reminders to re-experience trauma symptoms, with emotional numbing due to the individual's inability to respond appropriately [36].

A prior review and meta-analysis by Bao et al. [37] emphasized the functional alterations of the large-scale brain networks in PTSD populations. In detail, the authors reported hypoconnectivity within DMN and between the networks (DMN, SN) and significant associations of these brain networks with traumatic experiences. Additionally, hyperconnectivity between the large-scale brain networks (e.g., SN-SMN and DMN-SMN) was specifically associated with PTSD patients. Alterations in FC within or between the large-scale brain networks have been less frequently investigated to date among PTSD, trauma-exposed, and healthy participants. While the review by Bao et al. [37] provided valuable insights into network-level alterations, it was restricted by specific exclusion criteria. Specifically, Bao et al. [37] excluded studies utilizing non-seed-based approaches, such as regional homogeneity, and omitted research focused on predefined regions

of interest. Thus, a further investigation, including all functional imaging modalities and local connectivity indices, is needed. Indeed, although previous studies suggested that traumatic experiences can induce lasting changes in brain structure and function, the underlying neurobiological mechanisms of PTSD remain a critical area of ongoing investigation [38]. Analyzing resting-state activation could reveal alterations in brain activation and network functioning (segregation, integration, and their overall efficiency), and it would help to identify the consistent patterns of altered connectivity by also addressing the discrepancies and methodological limitations within the existing literature to inform future studies [39, 40]. Crucially, particular attention in comparing clinically diagnosed PTSD patients with healthy individuals exposed to traumatic events could provide a relevant contribution to disentangling aberrant resting-state functional connectivity (RS-FC) patterns associated with PTSD in contrast to the immediate and potentially evolving neural responses to a traumatic event in the absence of a PTSD diagnosis [39, 40].

The aim of this systematic review is to investigate the relationship between traumatic experiences and the brain activation at rest, characterizing alterations in resting-state activity in individuals with PTSD and trauma-exposed healthy controls. By synthesizing findings across different neuroimaging modalities, we aim to delineate the specific neural signatures of the disorder from general trauma-related alterations, potentially revealing compensatory mechanisms or early vulnerability factors. Furthermore, we aim to provide a comprehensive neurobiological characterization of PTSD by considering both intrinsic regional activity (e.g., metabolism and blood flow) to overcome the limitations of previous systematic reviews (e.g., Bao et al. [37]) and functional connectivity patterns. We also examine how recent emotional or cognitive engagement may modulate these resting-state patterns, as integrating these diverse designs offers a broader understanding of how the post-traumatic brain maintains or alters its intrinsic organization following trauma-relevant or cognitively demanding contexts.

2. Materials and methods

The present systematic review complies with the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) bibliographic research and data communication in systematic reviews and meta-analyses [41]. The PRISMA 2020 Main Checklist was used to ensure transparent and complete reporting of the methodology and results (see Supplementary materials).

2.1. Search strategy

The electronic databases PubMed (MedLINE), Scopus, and PsycINFO (ProQuest) were systematically searched from January 2003 to September 2023 for original references on brain correlates to traumatic experiences and PTSD. The present systematic review included articles that were published in English, in peer-reviewed journals, or cited in reference lists, and which had an abstract available to review. The keywords for research included all possible combinations of the following search terms adopted to the specific database: “Post-traumatic stress disorder”, “Traumatic event”, “Adverse event”, “Traumatic experience”, “Functional near-infrared spectroscopy”, “Functional magnetic resonance imaging”, “Positron emission tomography”, and “Single-photon emission computerized tomography” (detailed search strategy is reported in Table S1).

2.2. Study eligibility criteria

Studies were included in this systematic review if they met the following criteria: (1) involved only human participants; (2) were original articles written and published in peer-reviewed English-language journals; (3) employed quantitative assessments of post-traumatic stress; (4) included resting-state functional connectivity analyses. Moreover, to reduce potential confounding effects related to age-related neurobiological changes [42] and to enhance sample homogeneity and improve comparability across studies, the age range was set to 18–65 years.

The exclusion criteria were (1) studies involving experiments on animals; (2) studies conducted on children; (3) single-case studies; (4) qualitative studies; (5) books, reviews/non-original data, dissertations, conference presentations, and brief communications. Thus, to ensure sufficient methodological detail for consistent data extraction and synthesis, only peer-reviewed, full-length original articles were included.

2.3. Data screening

After reading titles and abstracts, the full texts of possibly relevant articles were read. Two reviewers (L.S. and S.I.) independently reviewed the studies for final inclusion or exclusion for qualitative synthesis of the results. A third independent reviewer (F.P.) was available for a decisive opinion in case the two reviewers disagreed.

2.4. Study selection

The study selection is described in the PRISMA flow chart (**Figure 1**). A total of 2435 unique articles were retrieved after the initial database search on PubMed (MedLINE), Scopus, and PsycINFO (ProQuest). Twenty-eight articles were excluded as duplicates, and 2249 articles did not match the eligibility criteria. Of these, 158 articles were assessed for eligibility for full-text reading, and 2 articles were manually added from citation searching. After full-text reading, 143 articles were excluded for methodological reasons (e.g., not a trauma-related study). A total of 17 original articles on RS-FC and PTSD were included in the qualitative synthesis of the studies in the present review.

2.5. Data extraction

Seventeen original articles were selected for qualitative synthesis. For each selected article, the following data were extracted: first author, publication year, country, and sample characteristics (sample size: number of healthy participants, patients with PTSD, trauma-exposed controls with or without PTSD, age, gender, and education). These data and labels are reported in **Table 1**. Study method, i.e., diagnostic criteria, brain imaging technique, comorbidity, time since traumatic events, task/stimuli, and main results, were also extracted and are reported in **Table 2**. A list of abbreviations of anatomical brain regions and terminologies of the methodological approaches in this article and specifically used in **Table 2** are reported in Table S2.

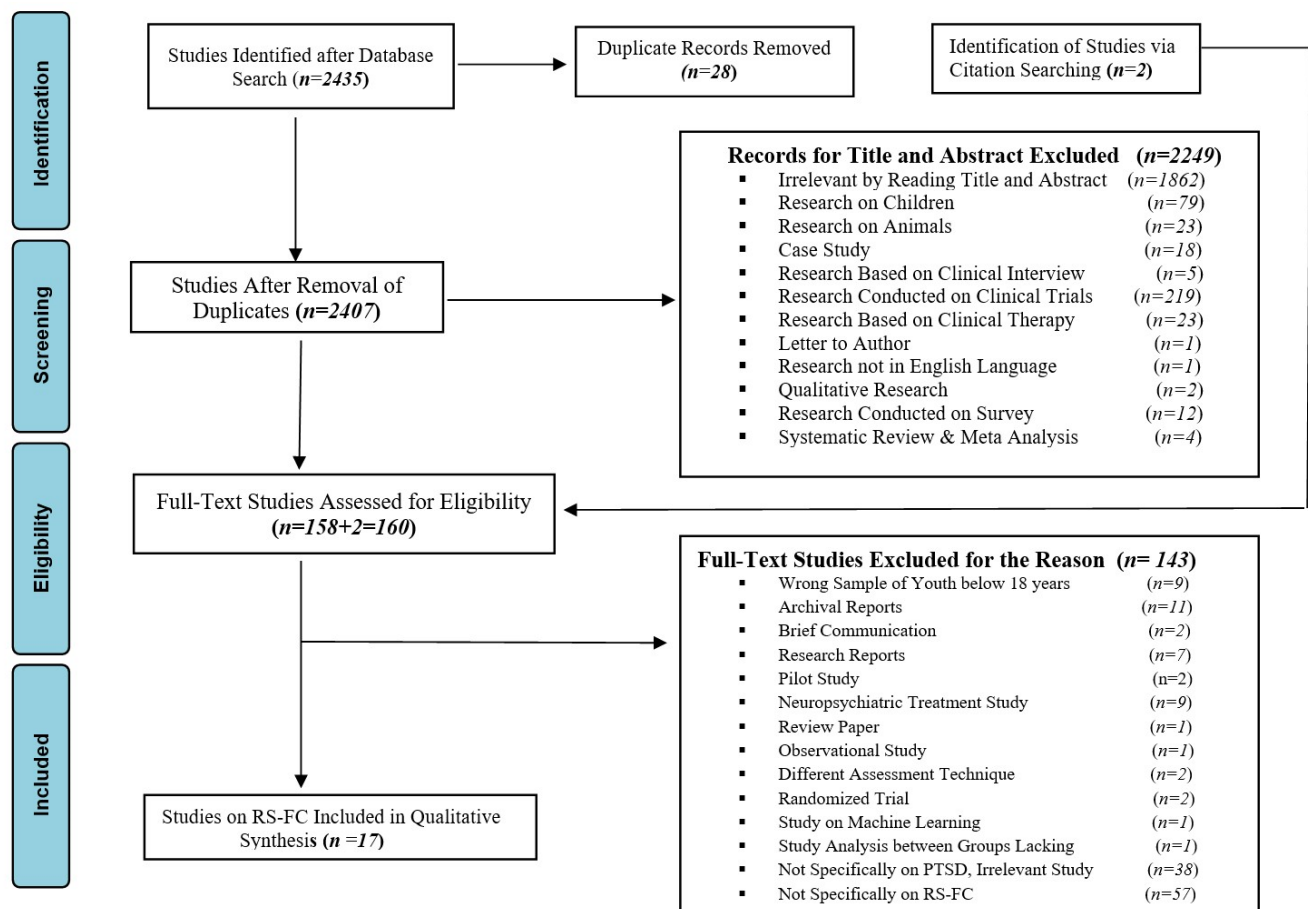


Figure 1 • Flow chart according to PRISMA guidelines. Post-traumatic stress disorder (PTSD), resting-state functional connectivity (RS-FC).

Table 1 • Author, country, and demographic characteristics of sample among studies included in the systematic review ($n = 17$).

Author	Country	Sample characteristics				
		Sample	Subsample	Mean age, years ($\pm$ SD)	Sex F/M	Education, mean ($\pm$ SD)
Chen et al. [43]	China	90	HP = 30 PTSD = 27 TEP = 33	HP = 49.9 ($\pm$ 6.1) PTSD = 48.4 ($\pm$ 10.3) TEP = 48.5 ($\pm$ 7.5)	HP = 7/23 PTSD = 7/20 TEP = 7/26	HP = 9.7 ($\pm$ 3.3) PTSD = 6.4 ($\pm$ 3.4) TEP = 7 ($\pm$ 3.4)
Clausen et al. [44]	USA	41	PTSD = 39	PTSD = 32.03 ($\pm$ 7.43)	Entire sample 39 M	NA
Dai et al. [45]	China	90	HP = 30 PTSD = 27 TEP = 33	HP = 49.9 ($\pm$ 6.1) PTSD = 48.4 ($\pm$ 10.3) TEP = 48.5 ($\pm$ 7.5)	HP = 23/7 PTSD = 20/7 TEP = 26/7	HP = 9.7 ($\pm$ 3.3) PTSD = 6.4 ($\pm$ 3.4) TEP = 7.0 ($\pm$ 3.4)
Fitzgerald et al. [46]	USA	90	TEP = 90	TEP = 18–50, 22.12 ($\pm$ 3.72)	Entire sample 90 F	NA
Fu et al. [47]	China	38	HP = 18 PTSD = 20	HP = 24–52, 42.17 ($\pm$ 8.38) PTSD = 25–52, 38.7 ($\pm$ 8.2)	HP = 11/7 PTSD = 13/7	HP = 9.5 ($\pm$ 3.7) PTSD = 11.2 ($\pm$ 2.7)
Gvozdanovic et al. [48]	Switzerland	73	Trauma film = 37 Control film = 36	Entire sample HP = 18–36, 23.4 ($\pm$ NA)	Entire sample 73 F	NA
Helpman et al. [49]	Israel	180	PTSD = 102 TEP = 78	NA	PTSD = 45/57 TEP = 40/38	NA
Jin et al. [50]	China	134	HP = 77 PTSD = 57	HP = 41.73 ($\pm$ 8.46) PTSD = 41.32 ($\pm$ 9.07)	HP = 53/24 PTSD = 43/14	HP = 7.55 ($\pm$ 2.81) PTSD = 7.77 ($\pm$ 2.98)
Ke et al. [51]	China	90	HP = 30 PTSD = 27 TEP = 33	HP = 49.9 ($\pm$ 6.1) PTSD = 48.4 ($\pm$ 10.3) TEP = 48.5 ($\pm$ 7.5)	HP = 23/7 PTSD = 20/7 TEP = 26/7	HP = 9.7 ($\pm$ 3.3) PTSD = 6.4 ($\pm$ 3.4) TEP = 7.0 ($\pm$ 3.4)
Kim et al. [52]	Republic of Korea	37	HP = 25 (HP-SPECT = 10) (HP-PET = 15) PTSD = 12	HP-SPECT = 26–50, 37.2 ($\pm$ 10.4) HP-PET = 32–53, 38.4 ($\pm$ 12.1) PTSD = 19–51, 35.9 ($\pm$ 13.8)	Entire sample 37 F	NA
Quidé et al. [53]	Australia Data Collection (France)	44	HP = 19 PTSD = 10 TEP = 15	HP = 18–53 PTSD = 18–44 TEP = 19–53	Entire sample 44 F	HP = 11–18 PTSD = 10–18 TEP = 11–18
Suarez-Jimenez et al. [54]	USA	54	HP = 16 PTSD = 17 TEP = 21	Entire sample: 18–80 HP = 37.13 ($\pm$ 10.30) PTSD = 45.71 ($\pm$ 15.32) TEP = 37.05 ($\pm$ 12.24)	HP = 9/7 PTSD = 8/9 TEP = 8/13	HP = 16.56 ($\pm$ 2.94) PTSD = 13.94 ($\pm$ 2.11) TEP = 15.29 ($\pm$ 1.93)
Sun et al. [55]	USA	26	PTSD = 11 TEP = 15	Entire sample: 28–55 43.5 ($\pm$ 8.8)	Entire sample PTSD = 2/24	NA
Suo et al. [56]	China	122	PTSD = 64 TEP = 58	Entire sample: 20–67 43.0 ($\pm$ 10.3)	Entire sample: 86/36	Entire sample: 0–16, 7.3 ($\pm$ 3.4)
Wang et al. [57]	China	94	PTSD = 59 TEP = 35	PTSD = 45.64 ($\pm$ 5.38) TEP = 44.47 ($\pm$ 9.73)	PTSD = 40/19 TEP = 29/6	PTSD = 8.19 ($\pm$ 3.24) TEP = 8.71 ($\pm$ 3.02)

Table 1 • Cont.

Author	Country	Sample characteristics				
		Sample	Subsample	Mean age, years ($\pm$ SD)	Sex F/M	Education, mean ($\pm$ SD)
Yin et al. [58]	China	126	PTSD = 54 TEP = 72	Entire sample: 18–59	NA	NA
Zhu et al. [59]	USA	50	PTSD = 17 TEP = 23	PTSD = 35.4 ($\pm$ 8.9) TEP = 35.3 ($\pm$ 10.4)	PTSD = 19/7 TEP = 17/7	PTSD = 14.4 ($\pm$ 1.9) TEP = 15.9 ($\pm$ 1.9)

Values indicated as means $\pm$ standard deviations: SD (parenthesis), healthy participants (HP), post-traumatic stress disorder (PTSD), trauma-exposed participants (TEP), not available (NA), Female/Male (F/M), healthy participants underwent positron emission tomography (HP-PET), healthy participants underwent single-photon emission computerized tomography (HP-SPECT). Education calculated in percentages not included. The information applies to the initial sample, after discarding some of the participants' information that was not provided.

Table 2 • Studies comparing methods, neuroimaging techniques, diagnoses, comorbidity, time since traumatic events, task/stimuli, and main results, and quality assessment ratings (Q) of included studies in the systematic review ($n = 17$).

Study	Imaging technique	Diagnoses	Comorbidity	Time since traumatic events	Research design/task/stimuli	Main relevant findings	Q
Chen et al. [43]	RS-fMRI	CAPS DSM-IV SCID DSM-IV	NA	NA	Between-Subject Design. The same selected group of participants from Ke et al. [51] was investigated. Typhoon survivors' intrinsic FC and brain functions were assessed by degree centrality and FC.	Increased DC in the left-PHG in PTSD and TEP groups as compared to HP. PCL-C scores were positively correlated with dynamic FC between PHG and left-SFG but negatively correlated with those between PHG and right-MFG.	5
Clausen et al. [44]	RS-fMRI FC	CAPS DSM-IV	NA	NA	Within-Subject Design RS-fMRI data acquisition was performed for the participants following fMRI scan.	Decreased FC in rostral ACC, MPFC (DMN), and right-anterior-lateral PFC (CEN) worsened PTSD symptoms.	6
Dai et al. [45]	RS-fMRI FC	CAPS: DSM-IV SCID DSM-IV	Depression	July 2014 typhoon survivors were screened in November 2014/January 2015. No analysis on distance of traumatic event.	Between-Subject Design. The sample was the same as Ke et al. [51] dFC was observed by RS-fMRI in DMN and cerebellum brain regions.	Clustering analysis identified two distinct dynamic FC states. State I (predominantly within-network) occurred more frequently in PTSD and TEP than in HP; however, the PTSD group exhibited weaker within-DMN connectivity compared to HP. State II (between-network) was characterized by reduced connectivity strength in PTSD patients, specifically involving DMN–cerebellum and DMN–VIS networks when compared to HP.	6
Fitzgerald et al. [46]	RS-fMRI FC	PCL-C	MDD GAD ADHD SUD AUD SAD AGP	NA	Correlational study. Whole-brain bilateral AMG FC maps were computed in participants exposed to traumatic events of natural disaster by 5 min RS-fMRI. The PCL-C was used to assess the severity of symptoms.	AMG FC is a significant predictor of PTSD severity. The brain regions that mostly predicted PTSD severity were the AMG, OFC, and dlPFC.	4

Table 2 • Cont.

Study	Imaging technique	Diagnoses	Comorbidity	Time since traumatic events	Research design/task/stimuli	Main relevant findings	Q
Fu et al. [47]	RS-fMRI FC	CAPS DSM-IV	NA	With reference to time post-trauma, participants were recruited within a definite time period. PTSD participants who received first aid in emergency after traffic accident were recruited between January 2016/October 2017.	Between-Subject Design. Drug-naïve PTSD patients who had experienced severe traffic accidents and matched healthy participants were imaged using RS-fMRI and analyzed by the algorithm of regional homogeneity (ReHo).	Decreased ReHo in the right-AG was observed in PTSD patients as compared to HP. Moreover, a significant negative correlation between ReHo in AG and CAPS scores was observed.	8
Gvozdanovic et al. [48]	RS-fMRI FC	NA	NA	Participants were scanned immediately after viewing experimental trauma film.	Within-Subject Design. RS-fMRI measurement based on two seeds (AMG and HIP) was conducted before and directly after the Trauma Film Paradigm.	Trauma exposure leads to rapid modifications in functional connectivity. Findings demonstrate decreased FC alterations associated with trauma in AMG and MTG and increased connectivity between HIP and precuneus in post-film RS-fMRI.	5
Helpman et al. [49]	RS-fMRI FC	CAPS: DSM-IV SCID DSM-IV	NA	NA	Between-Subject Design. To test sex-related patterns among trauma-exposed subjects with and without PTSD, RS-fMRI and a whole-brain seed-based analysis were used with key nodes in CEN (lPFC), SN (ACC, AMG, insula), DMN (MPFC), and HIP hubs as seed regions.	Compared to PTSD males and healthy trauma-exposed females, females with PTSD showed both weaker RS-FC between the AMG and PCC, and the precuneus and HIP, and stronger RS-FC between precuneus and ACC. Individuals with PTSD showed lower connectivity between the CEN-SN and CEN-DMN, within CEN, and within DMN, and higher connectivity in ACC-PCC pathway compared to TEP.	4
Jin et al. [50]	RS-fMRI FC	CAPS DSM-IV	NA	Earthquake survivors were recruited 8 months after an earthquake (8 May 2008).	Between-Subject Design. Patterns of whole-brain functional connectivity were investigated in post-earthquake individuals with or without PTSD.	PTSD patients had weak positive connectivity between MPFC, AMG, HIP, PHG, and rectus, as well as between inferior OFC and HIP, and strong negative connectivity between PCC and insula. CAPS scores were negatively correlated to the connectivity between AMG and MPFC in the patients with PTSD.	7
Ke et al. [51]	RS-fMRI FC	CAPS DSM-IV SCID DSM-IV	NA	July 2014 typhoon survivors were screened in November 2014/January 2015. No analysis for time post-event.	Between-Subject Design. MRI scans were conducted on typhoon-exposed participants with or without PTSD. Local and remote FC were assessed, as well as their association with clinical symptoms, by using ReHo- and ROI-based FC analyses.	PTSD patients had decreased ReHo in the left PCC/precuneus as compared to the trauma-exposed individuals and increased ReHo in the MCC/dACC relative to HP.	6

Table 2 • Cont.

Study	Imaging technique	Diagnoses	Comorbidity	Time since traumatic events	Research design/task/stimuli	Main relevant findings	Q
Kim et al. [52]	SPECT PET	NA	NA	NA	Between-Subject Design. Women suffering from PTSD as a result of experiencing sexual assault underwent brain perfusion SPECT and PET to study the resting cerebral blood flow (RS-CBF) and glucose metabolism. Two HP groups underwent only one of the imaging scans.	PTSD patients had significantly lower cerebral blood perfusion in the left-HIP and the left basal ganglia compared with the HP. The PTSD group also had lower cerebral glucose metabolic activity in the left-HIP, the left-STG and precentral gyri than the HP. Glucose metabolism was higher in PTSD group in the cerebelli compared to HP. PTSD patients exhibited lower uptake of radiotracers in the left hippocampal area compared with HP.	3
Quidé et al. [53]	RS-fMRI FC	CAPS: DSM-IV M.I.N.I.	NA	RS-fMRI data of 25 female participants 3 weeks (23 days) following their exposure to sexual assault were collected. After 6 months of trauma exposure, 10 participants diagnosed with PTSD were scanned again. TEP = 15 recovered from PTSD symptoms.	Between-Subjects Design. Intrinsic connectivity was assessed as a measure of network centrality by RS-fMRI. The presence of PTSD has been assessed 6 months after the scanning.	Increased FC was observed in PCC in PTSD patients in comparison to TEP and HP. Whereas, TEP participants had decreased FC in PCC and precuneus compared to HP.	6
Suarez-Jimenez et al. [54]	RS-fMRI FC	CAPS-V DSM-V SCID-V	NA	NA	Between-Subject Design. RS-FC fMRI scans were recorded. Participants then completed a free-viewing eye-tracking task outside the scanner assessing attention allocation. DT% on negatively valenced stimuli were assessed relative to FC.	Findings of group differences in attention allocation of negatively valenced stimuli showed PTSD patients and TEP had increased dwell time compared to HP. Higher DT% on negatively valenced stimuli was associated with fragile FC in NAcc-OFC brain regions in PTSD patients. In contrast, enhanced association was observed in HP. No significant difference was found on DT% between TEP and PTSD group.	6
Sun et al. [55]	RS-fMRI FC	CAPS: DSM-IV DSM-V	NA	Combat veterans who experienced PTSD symptoms in past month. No analysis of distance of traumatic event.	Correlational study. Self-reported measures including the two subscales of MIES (transgression and betrayal; Study 1) or the CAPS (Study 2).	Subjective appraisal of morally injurious events recorded by MIES index and PTSD symptoms were assessed by RS-FC and amplitude of low-frequency fluctuations. Findings reported an association of ALFF with L-IPL on MISE index. FC between L-IPL and precuneus was strongly associated with PTSD symptoms but negatively associated with MIES subscale scores.	5

Table 2 • Cont.

Study	Imaging technique	Diagnoses	Comorbidity	Time since traumatic events	Research design/task/stimuli	Main relevant findings	Q
Suo et al. [56]	RS-fMRI FC	CAPS DSM-IV SCID DSM-IV	NA	RS-fMRI scans were obtained from earthquake survivors 10–15 months later the traumatic event (2008 earthquake). No analysis of distance of traumatic event.	Correlational study. RS-fMRI scans were obtained from earthquake trauma survivors, and CPM technique was used to assess PTSD symptom severity.	PTSD symptom severity was assessed by CAPS. The association between CAPS-IV and FC was assessed by connectome-based model and LOOCV. Findings reported that FC can reliably predict PTSD patients' scores. The involvement of visual cortex, CB, and limbic and motor cortex was associated with the symptom severity of PTSD.	7
Wang et al. [57]	RS-fMRI FC	CAPS DSM-IV SCID. I/P DSM-IV-TR Axis I Disorders	MDD = 17 GAD = 2 PD = 1 DYS = 2	Survivors of 2008 earthquake recruited during January 2012 to October 2016. No analysis on distance of traumatic event.	Correlational study. Earthquake survivors (with and without PTSD) were scanned using RS-fMRI. FC seeds involved AMG and ACC, AMG and MPFC, CMA and HIP, and BLA and PCC bilaterally.	Sleep disturbance (PSQI) and trauma severity (CAPS) were negatively correlated with the RS-FC between the left BLA-left-MPFC and right BLA-right MPFC. BLA-PCC RS-FC has positive association with CAPS, CMA-HIP RS-FC was positively correlated with clinical measure of PSQI scores. PSQI was negatively associated with FC between BLA-MPFC.	6
Yin et al. [58]	RS-fMRI FC	CAPS DSM-IV PCL	NA	Earthquake survivors were recruited 10–15 months later the traumatic event (2008 earthquake). No analysis of duration since trauma event.	Between-Subject Design. RS-fMRI was recorded in earthquake survivors, comparing PTSD and TEP groups. Regional homogeneity (ReHo) was assessed.	PTSD group had enhanced ReHo in the L-IPL and right-SFG and reduced ReHo in the right-MTG and LG, relative to the TEP participants.	6
Zhu et al. [59]	RS-fMRI FC	CAPS DSM-IV SCID DSM-IV	PTSD Group MDD = 10 SOP = 2 PD = 1 SP = 1	PTSD participants were assessed in 10 sessions pre–post treatment scanning. TEP complete baseline assessment and then waited for 10 weeks.	Between-Subject Design. PTSD and TEP participants were assessed by RS-fMRI.	In post- vs. pre-treatment comparison, PTSD participants had significantly lower RS-FC in BLA and CMA-OFC and HIP-MPFC at pre-treatment than TEP; no significant group differences found for these pathways post-treatment. Increase in CMA-OFC pathways with RS-FC correlated with reduction in PTSD patients re-experiencing symptoms, and there was a negative correlation between increased BLA-vmPFC connectivity at resting state and reduced hyperarousal symptoms in PTSD.	5

Abbreviations of terms included in the table are: post-traumatic stress disorder (PTSD), major depressive disorder (MDD), generalized anxiety disorder (GAD), attention deficit hyperactivity disorder (ADHD), substance use disorder (SUD), social anxiety disorder (SAD), alcohol use disorder (AUD), panic disorder (PD), agoraphobia (AGP), specific phobia (SP), social phobia (SOP), dysthymia (DYS), clinician-administered PTSD scale (CAPS), diagnostic and statistical manual (DSM-IV/V), structured clinical interview for DSM (SCID), PTSD checklist-civilian version (PCL-C), functional connectivity (FC), parahippocampal gyrus (PHG), superior frontal gyrus (SFG), medial frontal gyrus (MFG), angular gyrus (AG), hippocampus (HIP), magnetic resonance imaging (MRI), functional MRI (fMRI), resting-state fMRI (RS-fMRI), anterior cingulate cortex (ACC), medial prefrontal cortex (MPFC), prefrontal cortex (PFC), central executive network (CEN), dynamic functional connectivity (dFC), posterior cingulate cortex (PCC), default mode network (DMN), default mode network-visual network (DMN-VIS), amygdala (AMG), dorsolateral prefrontal cortex (dlPFC), cerebellum (CB), basolateral amygdala (BLA), leave-one-out cross validation model (LOOCV), left inferior parietal lobule (L-IPL), dwell time (DT), amplitude of low frequency fluctuation (ALFF), lateral prefrontal cortex (LPFC), nucleus accumbens (NAcc), orbitofrontal cortex (OFC), dorsal anterior cingulate cortex (dACC), region of interest (ROI), ventromedial prefrontal cortex (vmPFC), salience network (SN), resting-state functional connectivity (RS-FC), centro-medial amygdala (CMA), degree of centrality (DC), healthy participants (HP), lingual gyrus (LG), regional homogeneity (ReHo), medial temporal gyrus (MTG), single-photon emission computerized tomography (SPECT), positron emission tomography (PET), cerebral blood flow (CBF), Moral Injury Events Scale (MIES), Mini-International Neuropsychiatric Interview (M.I.N.I.), trauma-exposed participants (TEP), Pittsburgh Sleep Quality Index (PSQI), not available (NA), medial cingulate cortex (MCC), connectome-based predictive modelling (CPM), superior temporal gyrus (STG).

2.6. Methodological quality assessment

Studies were reviewed for methodological quality assessment and sources of potential biases using a modified version of the Newcastle Ottawa Scale (NOS) [60] adapted to the methodology of the present study (Table S3). This scale evaluates the risk of selection bias, comparability of comparison groups, risk of potential confounders, and validity of outcome attainment for observational non-randomized investigations. Five judges (S.I., L.S., F.P., L.C., and R.D.B.) collaborated on the quality evaluation of the studies included in the present systematic review.

The NOS classifies each study as “very good” (9–10 stars), “good” (7–8 stars), “satisfactory” (5–6 stars), or “unsatisfactory” (0–4 stars) based on a star scoring system (Table S3).

3. Results

3.1. Synthesis of overall findings of included studies

The 17 included studies were based on convenient samples including healthy participants (HP: $n = 318$), participants with PTSD (PTSD: $n = 486$), and trauma-exposed participants (TED) without PTSD ($n = 491$), as described in **Table 1**. For studies involving patients, the PTSD diagnosis was based on the Clinician-Administered PTSD scale for Diagnostic and statistical manual DSM-IV [61] and DSM-V [62] (clinician-administered PTSD scale; CAPS: $n = 15$; DSM-IV: $n = 13$, DSM-5: $n = 2$).

Judges rated three studies as being of a good quality [47, 50, 56]. Eleven studies were rated as satisfactory [43–45, 48, 51, 53–55, 57–59]. Three studies were rated as unsatisfactory [46, 49, 52]. The main reasons for downgrading regarded the representativeness of the sample, unjustified sample size, and the omission of relevant methodological information (Table S4).

3.1.1. Research designs

Eleven studies adopted between-subject designs [43, 45, 47, 49–54, 58, 59], two studies adopted within-subject designs [44, 48], and four studies adopted a correlational [46, 55–57] design (**Table 2**).

Two studies employed novel analytical approaches, such as multi-voxel pattern analysis [46] and connectome-based predictive modeling [56].

3.1.2. Co-occurring disorders

Comorbidity of disorders was mentioned in four studies [45, 46, 57, 59]. In one study [59], PTSD participants presented comorbidity with depressive disorder, panic disorder, specific phobia, and social phobia.

In three studies [45, 46, 57], comorbidity of disorders was described over the entire sample. In the study by Wang et al. [57], the comorbidity of major depressive disorder, generalized anxiety disorder, panic disorder, and dysthymia was reported in the entire sample. In two studies [45, 46], the exact number of individuals suffering from comorbid disorders was not reported (**Table 2**).

3.1.3. Task/stimuli

Although RS-FC normally measures individual brain function in the absence of explicit cognitive tasks, three studies [44, 48, 54]

also included specific tasks/stimuli. A first study [54] involved negatively valenced faces presented during a free-viewing eye-tracking task to assess the relationship between RS-FC and the time spent on negatively valenced stimuli. In another study [48], functional connectivity changes at rest were recorded directly after a trauma-related or a control movie. In a further study [44], participants performed a multisource interference task (MSIT) with congruent–incongruent conditions in order to identify the seeds for subsequent resting-state connectivity analyses.

3.2. Neuroimaging results

The included studies employed RS-fMRI ($n = 16$) except for one study [52] on positron emission tomography (PET) and single-photon emission computerized tomography (SPECT), which utilized brain perfusion, resting-state cerebral blood flow, and glucose metabolism (**Table 2**).

The brain regions that were most implicated in the primary outcomes of the included studies were found to be cortical regions, such as the PFC and the cingulate cortex, and subcortical regions, such as the AMG and the HIP (**Table 2**).

The alterations encompass changes in regional spontaneous activity, functional connectivity (both static and dynamic), and metabolic patterns [43, 50, 52–54, 56, 57], and then delve into the results of the large-scale networks.

3.3. Key brain regions

3.3.1. Amygdala

Eight studies reported significant alterations of the AMG in individuals with traumatic experiences and PTSD [46, 48–51, 55, 57, 59]. PTSD patients exhibited changes in ReHo in the AMG when compared to both trauma-exposed controls and healthy controls [51].

With respect to functional connectivity, studies reported functional alterations in several areas. Both the RS-FC between the left basolateral amygdala (BLA) and left-MPFC and between right BLA and right-MPFC were associated with PTSD symptoms and poor sleep quality in trauma-exposed individuals [57]. Furthermore, FC between the right-AMG seed and right thalamus was negatively correlated with moral injury betrayal scores [55].

In PTSD patients, as compared to trauma-exposed non-PTSD controls (TEP), decreased FC between the right-AMG and left-MPFC was observed. This pattern is negatively correlated with CAPS scores [50]. Increased amygdala RS-FC with the orbitofrontal cortex (OFC) was also found in patients with PTSD [59].

Immediate experimental trauma exposure led to decreases in connectivity between the AMG and the middle temporal gyrus in healthy subjects [48]. Moreover, greater AMG-FC with the caudate, cerebellum, superior parietal cortex, PCC, and supra-marginal gyrus predicted higher PTSD symptoms [46]. However, one study [57] noted a negative correlation for AMG–PCC with CAPS scores, highlighting existing inconsistencies in the literature.

Considering the sex differences, the amygdala–precuneus pathway exhibited significant group-by-sex interaction effects in trauma-exposed individuals, with females with PTSD showing weaker RS-FC than PTSD males and healthy trauma-exposed

females, thus indicating that connectivity patterns in PTSD can differ based on sex [49].

3.3.2. Hippocampus and parahippocampal gyrus

Six studies reported alterations of the hippocampus [48–51, 57, 59] or the parahippocampal gyrus [43, 47, 50, 51].

The PTSD patients demonstrated increased ReHo in left-HIP and bilateral parahippocampal gyrus (PHG) [51]. Moreover, positive correlations were found between PTSD symptom severity (PCL scores) and ReHo in the PHG across all traumatized subjects [51].

Significant relative decreases in perfusion and glucose metabolic activity in the left hippocampus were observed in women with acute PTSD related to sexual assault compared to controls [52]. These changes are thought to be linked to memory deficiencies and the process of fear extinction [52].

As regards the functional connectivity, findings by Jin et al. [50] reported that PTSD patients had weak positive connectivity between the MPFC, AMG, HIP, PHG, and rectus, as well as between the inferior OFC and HIP, and strong negative connectivity between the PCC and insula [50]. Furthermore, patients with PTSD showed reduced RS-FC between the HIP and the ventromedial PFC (vmPFC) as compared with the trauma-exposed healthy control group at pre-treatment [59]. Additionally, trauma-related increases in functional connectivity between the hippocampus and precuneus were observed immediately after experimental trauma in healthy subjects [48].

The centro-medial amygdala–hippocampus (CMA-HIP) RS-FC was positively correlated with Pittsburgh Sleep Quality Index (PSQI) scores in trauma-exposed individuals [57]. Chen et al. [43] reported an increased degree of centrality (DC) in the left-PHG in both the PTSD group and the trauma-exposed control group relative to healthy controls. Moreover, both PTSD patients and trauma-exposed individuals had decreased dynamic functional connectivity (dFC) among left-PHG, bilateral medial frontal gyrus (MFG), and superior frontal gyrus (SFG) as compared to HP [43].

Related to the sex differences, the HIP–precuneus pathway exhibited significant group-by-sex interaction effects, with females with PTSD showing lower connectivity between the HIP and precuneus compared to trauma-exposed controls [49].

3.3.3. Prefrontal cortex

Several articles reported alterations of the prefrontal cortex and its subregions in PTSD, and some of these have already been described in the above sections.

PTSD patients showed ReHo changes in the dlPFC relative to control groups [51]. Increased ReHo in the insula/inferior frontal gyrus was observed in typhoon survivors compared to healthy controls [51].

Furthermore, enhanced ReHo in the right SFG was found in PTSD patients relative to controls [58]. Moreover, altered ReHo in this brain area was correlated with the severity of clinical symptoms [58]. Jin et al. [50] also reported weak positive connections between the left-SFG and the right hippocampus.

Concerning the medial cortex, Clausen et al. [44] reported decreased activation and FC in the MPFC and rostral anterior cingulate cortex (rACC) during cognitive inhibition (interference processing) followed by increased activation in the congruent trials, which were associated with PTSD symptom severity. Worse PTSD symptoms and neuropsychological performance were associated with decreased FC between two brain networks: the DMN (MPFC) and CEN (IPFC). The study findings clearly indicate that PTSD symptoms and neuropsychological performance were associated with the dysregulation of MPFC, rACC activation, and FC patterns [44].

3.3.4. Cingulate cortex

Ke et al. [51] reported that PTSD patients had decreased ReHo in the left-PCC/precuneus as compared to the trauma-exposed individuals and increased ReHo in the medial cingulate cortex (MCC) and dACC relative to healthy participants.

In the comparison of controls (TEP, HP), the trauma-exposed participants reportedly exhibited increased ReHo in the insula, IFG, and middle and dorsal anterior cingulate cortex (MCC/dACC) than in HP [51]. Additionally, enhanced negative FC was observed in TEP participants between the (MCC/dACC) and (PCC/precuneus) compared to HP.

Furthermore, a positive correlation was observed between ReHo and PTSD symptom severity in several brain regions of PTSD patients [51].

Clausen et al. [44] also discussed lower FC in the cingulate cortex and reported that rACC and left IFG were associated with worse neuropsychological performance [44]. Moreover, decreased intrinsic functional connectivity in the PCC/precuneus was also observed in trauma-exposed individuals compared to HP (but not in those with PTSD), potentially serving as a marker of resilience to trauma exposure [53].

Considering sex differences, females with PTSD had stronger RS-FC between the precuneus and anterior cingulate cortex than males with PTSD and healthy trauma-exposed females [49].

3.3.5. Cerebellum

Kim et al. [52] reported that glucose metabolism was higher in subjects with PTSD than in control subjects in both hemispheres of the cerebellum.

Suo et al. [56] explicitly classify the cerebellum as a macroscale brain region and, when parcellating nodes into canonical networks, it is grouped into the subcortical–cerebellum network. This study found that the positive network (functional connections predicting PTSD symptom severity) primarily utilized functional connectivity among the interconnected brain regions, such as the subcortical and cerebellum network, visual association network, and motor network. In particular, the bilateral visual cortex and cerebellum have critical roles in predicting the severity of PTSD-related symptoms, as reflected in CAPS-IV scores.

Individuals with PTSD demonstrated weaker between-network connection strength in DMN–cerebellum and DMN–VIS [45]. This finding is in line with those of Chen et al. [43], who reported

a decreased dynamic functional connectivity between the right-cerebellum and left-PHG in PTSD patients relative to healthy controls.

3.3.6. Insular cortex

The insula, a region of the SN, showed a specific pattern of functional alteration in PTSD.

Ke et al. [51] found that PTSD patients have increased ReHo in the right insula/IFG compared with both trauma-exposed individuals and healthy controls. Compared with the healthy controls, the trauma-exposed individuals also showed increased ReHo in the bilateral insula/IFG. Additionally, Jin et al. [50] compared PTSD patients with non-PTSD controls and observed strong negative FC between the bilateral insula and PCC in PTSD patients. Furthermore, Sun et al. [55] reported that the amplitude of low-frequency fluctuation in the right posterior insula was negatively related to subscales assessing moral injury.

3.3.7. Default mode network (DMN), salience network (SN), and central executive network (CEN) connectivity

The contribution of the areas that are part of the DMN (particularly, PCC, MPFC, precuneus, and ACC), SN (hippocampus and AMG), CEN (IFG and lateral prefrontal cortex), and ventral attention network (VAN, dACC) has already been described in previous sections. In addition, Fu et al. [47] demonstrated that compared to healthy controls, PTSD patients showed decreased ReHo values in the right angular gyrus (a key region of the DMN). Moreover, they found a significant negative correlation between the angular gyrus ReHo and the CAPS score.

Helpman et al. [49] has discussed decreased functional connectivity between CEN-SN and CEN-DMN. The authors found intrinsic functional connectivity within the CEN and DMN in individuals with PTSD as compared to healthy participants.

By means of a clustering analysis, Dai et al. [45] clustered all the FC matrices to identify reoccurring FC states, revealing two re-occurring functional connectivity “states” across all participants: State I (more frequent: 79%) was characterized by more strongly connected within-network connections, and State II (less frequent: 21%) was more strongly interconnected in comparison. State I focused on the within-brain networks’ FC, with abnormalities in dFC being significant in PTSD patients. In contrast, in State II, FC patterns were observed between the brain networks. The findings reported dFC alterations between the DMN–cerebellum and DMN–visual networks in PTSD patients. The between-network connection CEN-SN was stronger in the PTSD individuals than the controls.

4. Discussion

The present systematic review primarily aimed to explore the neural underpinnings of PTSD and related trauma exposure by synthesizing findings from studies utilizing resting-state fMRI. The review incorporated research on individuals with and without PTSD following diverse traumatic events, including natural disasters (e.g., typhoons), combat, and traffic accidents. The included studies collectively sought to identify neural markers for PTSD, predict symptom severity, and differentiate PTSD from other trauma-related conditions, such as moral injury.

The existing literature presents both consistent and inconsistent findings regarding brain alterations in PTSD. While studies consistently point to widespread alterations in intrinsic brain activity and FC in individuals with PTSD, e.g., [46, 56, 57], the precise nature (increased vs. decreased connectivity) and specific anatomical locations of these alterations often vary. This may be due to the limitations of the included studies and to the heterogeneous study designs, methodological approaches, different trauma types, diverse comparison groups, and often small sample sizes [43, 47, 51, 52].

Despite these inter-study differences, converging findings suggest that PTSD is associated with a distributed network of brain alterations rather than being confined to a single region [46]. Findings from the included studies highlight the relevance of brain areas extending beyond the classic traditional fear neural circuit and emphasizing broader network involvement [63, 64], including regions such as the caudate, cerebellum, superior parietal cortex, PCC, and supramarginal gyrus [46]. This expanding focus on non-traditional fear circuit regions, coupled with insights from models like the triple network model, is crucial for understanding symptom severity [56].

The DMN, most active during rest and involved in self-referential thought and episodic memory, shows significant alterations in connectivity in PTSD [43–45, 50, 53]. Indeed, disrupted FC within the DMN is considered a significant biomarker for classifying PTSD patients [45]. Altered FC of the medial prefrontal cortex, a core DMN hub, is a recurring finding [44, 50, 51, 57]. Specifically, decreased connectivity between the MPFC and HIP, as well as the parahippocampal gyri, has been observed in PTSD patients [50]. Interestingly, both PTSD groups and trauma-exposed control individuals exhibit increased DC in the left-PHG compared to healthy controls [43], suggesting that some alterations may be linked to trauma exposure itself. Moreover, reduced perfusion and glucose metabolic activity in the left hippocampus reported in women with acute PTSD are thought to be linked to memory deficiencies and the process of fear extinction [52].

Atypical AMG FC is widely considered a cardinal feature of PTSD [16, 65]. AMG FC patterns associated with PTSD include connections with the OFC, dlPFC, insula, and hippocampus [46, 50, 57, 59]. Specifically, the decreased FC between the AMG and the left-MPFC [50] and increased RS-FC with the OFC [59] have been reported in PTSD patients. Beyond established PTSD, immediate experimental trauma exposure can also lead to altered amygdala connectivity. For instance, decreases in connectivity between the AMG and the middle temporal gyrus were reported in healthy subjects after experimental trauma exposure [48]. Moreover, inconsistencies persist regarding the relationship between amygdala FC and symptom severity [15]; while Fitzgerald et al. [46] reported that greater AMG FC with certain regions, including the PCC, predicted symptom severity, Wang et al. [57] found a negative correlation for AMG–PCC connectivity with CAPS scores, highlighting existing discrepancies in the literature.

Regarding sex differences, the AMG–precuneus pathway exhibited significant group-by-sex interaction effects in trauma-exposed individuals. Females with PTSD showed weaker RS-FC than PTSD males and healthy trauma-exposed females, indicating that connectivity patterns in PTSD can differ based on sex [49]. However, despite sex being a crucial biological factor influencing

neural connectivity patterns in PTSD, only one included study by Helpman et al. [49] specifically assessed its effect on brain alterations following traumatic events. Prior research on the neural correlates of PTSD pathophysiology has commonly treated sex as a potential confounder, thus neglecting to examine sex-dependent patterns of RS-FC in PTSD.

A further study assessed the effect of therapeutic interventions, specifically exposure-based therapy (PE), on resting-state neural networks. While patients with PTSD exhibited significantly lower RS-FC between the AMG and the OFC and between the HIP and the vmPFC at pre-treatment compared to trauma-exposed healthy controls (TEP), no significant group differences were found at post-treatment, suggesting identifiable neuroanatomical targets for treatment [59].

Importantly, several studies highlighted that trauma exposure itself can induce brain functional changes, even in the absence of a PTSD diagnosis. These alterations may be observed very early after the traumatic event, suggesting potential early predictive biomarkers and the need for early preventive interventions [43, 48, 51, 53].

It is important to note that early intervention or maltreatment of post-trauma events might potentially worsen the symptoms associated with PTSD. Hence, possible neurobiological markers—if validated—should not automatically imply immediate intervention but rather inform careful risk monitoring and evidence-based decision-making [66–68].

More recent clinical guidelines, e.g., [69–71], have also emphasized that early interventions should follow established recommendations and avoid non-indicated or overly intensive approaches in the acute post-trauma phase.

The time interval between trauma exposure and neuroimaging assessment is a relevant variable that may influence resting-state connectivity findings [13]. Nevertheless, only two studies included in the present review specifically reported the role of the time interval between trauma exposure and neuroimaging acquisition. Quidé et al. [53] assessed participants three weeks after trauma exposure and again at follow-up after 6 months, allowing for the evaluation of longitudinal changes. Similarly, Zhu et al. [59] included repeated assessments in the context of treatment, indirectly addressing temporal effects. Investigations of long-term intervals (e.g., several years post-trauma) are scarce in the studies included.

A significant finding of our review is the heterogeneous reporting of psychiatric comorbidity across the included studies. Conditions such as depression and various anxiety disorders were frequently present [72, 73]. Studies included in this systematic review failed to provide information on comorbidity (except for four studies: [45, 46, 57, 59]) and did not conduct analyses accounting for the comorbidity. This poses a methodological challenge, as comorbid depressive or anxious symptoms may confound resting-state patterns and potentially obscure neurobiological markers specific to PTSD [74–76]. This issue is clarified further in the section on limitations.

Three studies [43, 45, 51] included in the present systematic review were found to include the same sample population (typhoon-exposed participants) reporting different data analyses and results.

The first study conducted by Ke et al. [51] reported changes in regional homogeneity in brain regions such as the AMG, parahippocampal gyrus, and prefrontal cortex, with CAPS scores positively correlated with ReHo in the left PCC/precuneus in participants with PTSD or TEP. This study was replicated by Dai et al. [45] who investigated the same selected group of participants reporting abnormalities in dynamic functional connectivity within the DMN in PTSD participants; particularly weaker connections with the cerebellum and visual areas were also observed. Finally, Chen et al. [43] assessed the degree centrality and seed-based FC, reporting an increased DC in the left-PHG in PTSD and TEP groups as compared to HP. The PTSD checklist-civilian version (PCL-C) scores were positively correlated with dynamic FC between PHG and left-SFG but negatively correlated with those between PHG and right-MFG.

Several limitations should be acknowledged in this systematic review. The first limitation concerns the specificity of the search terms employed. Indeed, the wide variation in terminology within the trauma literature—including the use of hyphenated versus non-hyphenated forms (e.g., ‘post-traumatic’ vs. ‘posttraumatic’) and the exclusive use of the ‘PTSD’ acronym—may have influenced the initial pool of captured studies. Although database-specific mapping algorithms were relied upon to minimize this risk, the potential for terminology-based selection bias remains an inherent limitation in this systematic review.

Furthermore, although fMRI is the primary tool for measuring brain activation and FC, PET/SPECT can measure resting-state metabolism and regional blood flow [77–79]. The reported alterations in regional cerebral blood flow (rCBF) provide a metabolic foundation that complements the fMRI-based FC findings. However, this systematic review only included one study using PET/SPECT [52], which allows for a preliminary cross-modal comparison.

The methodological heterogeneity of the included studies may have affected the results. Indeed, differences in trauma type, chronicity, and timing of exposure may reasonably contribute to discrepant neuroimaging findings [80]. Indeed, the studies included in our review encompass heterogeneous trauma populations (e.g., combat-related trauma, interpersonal violence, traffic accidents, and natural disasters). These experiences differ in terms of duration, repetition, interpersonal involvement, and developmental timing, all of which may differentially affect brain activation and large-scale brain networks implicated in threat processing, emotion regulation, and self-referential functioning. For example, prolonged and repeated combat exposure may be associated with sustained alterations in salience and fear-processing networks [64, 81], whereas interpersonal trauma may more specifically affect amygdala–prefrontal circuitry involved in threat appraisal and emotional regulation [82]. In addition, variability in sample size and analytic approaches may further contribute to inconsistencies across studies. The presence of co-occurring psychiatric disorders, e.g., depression and anxiety [75], and psychotropic medication use can also confound results. While some studies explicitly mentioned comorbidity [45, 46, 57, 59], other studies either did not report co-occurring mental disorders, e.g., [52], or considered the presence of psychiatric disorders other than PTSD as an exclusion criterion, e.g., [56], sometimes specifically including depression [49, 59]. A further limitation is that the present study is restricted to a systematic qualitative synthesis. While we aimed for maximum transparency, a

meta-analysis was not feasible due to the substantial variability in neuroimaging methodologies and reporting standards across the literature. Future research with more standardized protocols will be essential to permit a quantitative meta-analysis of these functional connectivity patterns. Moreover, the age limit of 65 years minimized the confounding influence of age-related neurobiological changes. However, at the same time, this age restriction limits the study's generalizability. PTSD is a chronic condition that persists into adulthood, and traumatic events can occur at any stage of life [42]. As the aging brain may exhibit different compensatory mechanisms or vulnerability factors, future research should specifically target this population to determine whether the identified neural signatures remain consistent throughout the lifespan. A further limitation concerns the variable methodological quality of the included studies. Only a small number of studies were rated as 'good,' while most were 'satisfactory' and three were 'unsatisfactory.' This pattern requires the results to be interpreted with caution. Specifically, the findings derived from lower-quality studies might be more susceptible to bias, underscoring the need for future PTSD neuroimaging research to adopt more rigorous standardized protocols and larger, more representative samples.

Finally, the literature search for this systematic review was finalized in 2023. We acknowledge that several significant studies—such as [16, 83–85]—have been published since the completion of our primary search. Nevertheless, a full systematic update was not feasible due to the rigorous requirements of re-screening and quality assessment. However, a preliminary observation of these recent works suggests that their findings are largely consistent with the patterns identified in our synthesis, reinforcing the stability and current relevance of our conclusions.

4.1. Implications and future directions

The findings from this systematic review have important implications for understanding PTSD and developing targeted interventions.

The utility of novel analytical approaches, such as multi-voxel pattern analysis [46] and connectome-based predictive modeling [56], has been demonstrated to predict individual PTSD symptom severity. These methods can help to characterize more accurately the neural underpinnings of PTSD and identify specific brain features related to illness severity [46, 56]. The presence of early, rapid, post-trauma functional connectivity changes highlights the critical need for early interventions to potentially reduce maladaptive effects of traumatic events on the brain [48].

The current evidence is not sufficient per se to support the routine use of functional neuroimaging for diagnosis or treatment monitoring in standard clinical settings. However, resting-state neuroimaging findings might provide support in the diagnostic process and may currently be considered primarily within a research framework. For instance, a study by Zhang et al. [86] focused on machine learning to use MRI to improve PTSD diagnoses.

As classification features, the studies used gray matter volume, amplitude of low-frequency fluctuation, and regional homogeneity. Their results demonstrated that using the combination of these features determined a higher accuracy in classifying the three groups. These findings support the use of a multimodal MRI approach for assisting clinical assessment.

In the longer term, validated neurobiological markers may complement psychological assessment tools, contribute to risk stratification after trauma exposure, and potentially inform the personalization of treatment. However, further large-scale and prospective studies are needed before such applications can be implemented in routine clinical practice.

Moreover, future research should prioritize longitudinal studies in at-risk populations to ascertain whether observed brain abnormalities represent pre-existing risk factors or are acquired deficits following trauma, thereby clarifying their nature as stable traits or transient states. Finally, further studies should focus on specific therapeutic interventions to assess their association with altered connectivity patterns, moving towards more targeted and effective neurobiologically informed treatments.

Finally, since PTSD is recognized as a significant risk factor for cardiac conditions (such as Takotsubo Cardiomyopathy [87]) and dementia [88–90], identifying neuroimaging biomarkers that track the long-term impact of traumatic stress is crucial. Specifically, understanding how altered brain–heart interactions and resting-state network dysfunctions contribute to the progression from post-traumatic stress to cognitive decline remains a key challenge. Subsequent investigations should determine whether specific alterations in these networks can serve as early predictors for dementia and other stress-related systemic pathologies in trauma-exposed populations.

5. Conclusions

This systematic review provides a comprehensive overview of the brain correlates of PTSD and trauma exposure, primarily utilizing advanced neuroimaging techniques. The findings underscore that PTSD is characterized by widespread alterations in functional brain networks, extending beyond the traditionally emphasized fear circuits to involve a broader set of regions critical for emotion regulation, self-referential processing, and cognitive control. While inconsistencies in the precise localization of connectivity changes persist across studies, largely due to methodological heterogeneity and diverse populations, the existing evidence points to significant alterations within the DMN, atypical AMG connectivity, and broader network disruptions. Crucially, this review highlights that brain alterations can manifest early after trauma exposure, even in the absence of a PTSD diagnosis, emphasizing the potential for early biomarkers and preventive strategies. Furthermore, the role of biological factors such as sex warrants greater consideration in future research to elucidate nuanced connectivity patterns. Ultimately, understanding these complex neural signatures is pivotal for developing more precise diagnostic tools and effective, neurobiologically informed therapeutic interventions for individuals affected by traumatic stress.

Acknowledgments

The authors acknowledge the use of Gemini AI (Google; version 2.5 pro) and DeepL (version 26.2.1) for language editing. The tools were applied to enhance clarity and scholarly tone, and all edits were carefully reviewed and validated by the authors to ensure

accuracy, compliance with academic standards, and research integrity. The authors fully support Academia.edu Journals' adherence to COPE guidelines on AI in publication ethics and confirm that this use has been managed responsibly and ethically.

Funding

This research received no external funding. The three authors, R.D.B., L.C., and F.P. were sponsored by grants from a EU-funded project sponsored by the NextGenerationEU-MUR M4C2.I.1.3 PNRR Extended Partnership initiative on Emerging Infectious Diseases (PE00000007, INF-ACT) "One Health Basic and Translational Research Actions addressing Unmet Needs on Emerging Infectious Diseases" under the grant INF-ACT Cascade Open Call 2023 (COC-1-2023-ISS-02)—CUP I83C22001810007—BEHAVE—MOD.

Author contributions

Conceptualization, S.I. and L.S.; methodology, S.I., L.S. and F.P.; software, S.I.; validation, L.S. and F.P.; formal analysis, S.I., F.P. and L.S.; investigation, S.I.; resources, S.I.; data curation, S.I., L.S. and F.P.; writing—original draft preparation, S.I., L.S. and F.P.; writing—review and editing, L.C. and R.D.B.; visualization, S.I.; supervision, L.S.; project administration, L.S. and S.I.; funding acquisition, F.P., L.C. and R.D.B. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

The authors declare that they have no competing interests. The sponsors played no role in the conception, execution, interpretation or writing of this manuscript.

Data availability statement

No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Supplementary materials

The Supplementary materials are available at <https://doi.org/10.20935/AcadNeurosci8253>. Reference [91] is cited in the Supplementary materials.

Additional information

Received: 29 December 2025

Accepted: 7 April 2026

Published: 29 April 2026

Academia Neuroscience and Brain Research papers should be cited as *Academia Neuroscience and Brain Research* 2026, ISSN 3071-4087, <https://doi.org/10.20935/AcadNeurosci8253>. The journal's official abbreviation is *Acad. Neurosci. Brain Res.*

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